

Note: $\vec{n} \parallel \frac{\partial \mathbf{x}}{\partial t} \times \frac{\partial \mathbf{x}}{\partial u} \parallel = \frac{\partial \mathbf{x}}{\partial t} \times \frac{\partial \mathbf{x}}{\partial u}$ by def'n

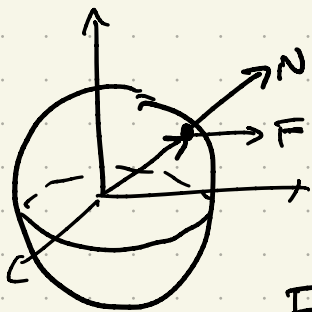
$$\text{so } \boxed{\iint_S \mathbf{F} \cdot \vec{n} \, d\sigma = \iint_R \mathbf{F}(\mathbf{x}(t, u)) \cdot \left(\frac{\partial \mathbf{x}}{\partial t} \times \frac{\partial \mathbf{x}}{\partial u} \right) dt du}$$

Often called "the flux of $\mathbf{F}$ through S "

Ex: $\mathbf{F}(x, y) = (x, y, 0)$, $S = \{x^2 + y^2 + z^2 = a^2\}$
w/ outward orientation

$$\mathbf{N}(\varphi, \theta) = \frac{\partial \mathbf{x}}{\partial \varphi} \times \frac{\partial \mathbf{x}}{\partial \theta} = a \sin \varphi \, \mathbf{x}(\varphi, \theta)$$

since $\varphi \in [0, \pi]$, $\mathbf{N}(\varphi, \theta)$ points in the direction
of the position vector $\mathbf{x}(\varphi, \theta)$ ($\sin \varphi \geq 0$)



$$\mathbf{x}(\varphi, \theta) = (a \sin \varphi \cos \theta, a \sin \varphi \sin \theta, a \cos \varphi)$$

$$\mathbf{F}(\mathbf{x}(\varphi, \theta)) = (a \sin \varphi \cos \theta, a \sin \varphi \sin \theta, 0)$$

$$\mathbf{F}(\mathbf{x}(\varphi, \theta)) \cdot \mathbf{N}(\varphi, \theta)$$

$$= (a \sin \varphi) (a^2 \sin^2 \varphi) = a^3 \sin^3 \varphi$$

$$\iint_S \mathbf{F} \cdot \vec{n} \, d\sigma = a^3 \int_0^{2\pi} \int_0^\pi \sin^3 \varphi \, d\varphi \, d\theta$$

$$= 2\pi a^3 \int_0^\pi \sin^3 \varphi \, d\varphi$$

$$= 2\pi a^3 \left(-\cos \varphi + \frac{1}{3} \cos^3 \varphi \right) \Big|_0^\pi = 2\pi a^3 \frac{4}{3} = \frac{8\pi a^3}{3}$$